NITROFURANTOIN

Avoid if eGFR <45 (or <30)

Lower UTI only

CHEMICAL NAME 1-[(5-nitrofuran-2-yl)methylideneamino]imidazolidine-2,4-dione
CLASS Nitrofuran urinary antiseptic
TRADE NAMES Macrobid · Macrodantin · Furadantin · Uvamin

THERAPEUTIC USES / INDICATIONS

- Uncomplicated lower UTI (cystitis) in women.
- Prophylaxis of recurrent UTI.
- UTI in pregnancy (avoid at term).

ADMINISTRATION

- **PO** with food or milk (↑ absorption, ↓ nausea).
- Swallow capsules whole.
- Suspension: do not mix with other liquids.

DOSAGES

- **Acute cystitis:** MR 100 mg every 12 h, or 50 mg every 6 h, for 3–7 days.
- **Prophylaxis:** 50–100 mg at night.
- **Children (>3 months):** 3 mg/kg/day divided every 6 h; prophylaxis 1 mg/kg at night.

PHARMACOKINETICS

- **Peak:** 30 min. Rapidly excreted → therapeutic levels only in urine (not blood or kidney tissue).
- Partly metabolised in tissues.
- Excreted in urine (turns dark yellow-brown). **t½:** 20–60 min.

PHARMACODYNAMICS

- Reduced by bacterial enzymes to reactive intermediates → damage ribosomes, DNA, cell wall → bactericidal in urine.
- Active against *E. coli*, *Enterococcus faecalis*, *Staph. saprophyticus*; not *Proteus* or *Pseudomonas*.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, anorexia.
- Brown discoloration of urine.
- Headache, dizziness.

ADVERSE EFFECTS (SERIOUS)

- Acute pulmonary reaction (fever, cough, dyspnoea) and chronic pulmonary fibrosis (long-term).
- Peripheral neuropathy (renal impairment).
- Hepatotoxicity (chronic hepatitis).
- Haemolytic anaemia (G6PD deficiency, neonates).

CONTRAINDICATIONS & CAUTIONS

- eGFR <45 mL/min (short courses may be used down to 30 in some guidelines) — ineffective and toxic.
- G6PD deficiency; acute porphyria.
- Infants <3 months; pregnancy at term (38–42 weeks) and during labour.
- Pyelonephritis or systemic infection (no tissue levels).

KEY INTERACTIONS

- **Magnesium trisilicate antacids:** ↓ absorption.
- **Probenecid:** ↓ urinary levels, ↑ toxicity.

NURSING MANAGEMENT

- Check renal function before starting.
- Give with food.
- Monitor for cough, breathlessness, chest pain.
- Long-term prophylaxis: monitor LFTs and lung function.
- Assess urinary symptoms; send MSU culture.

PATIENT EDUCATION

- Take with food or milk.
- Urine may turn dark yellow or brown — harmless.
- Report cough, breathlessness, numbness or yellow skin.
- Complete the course; drink fluids.

KEY POINTS

- Urinary antiseptic; cystitis only.
- Needs good renal function.
- Pulmonary reactions, neuropathy.
- Avoid at term (neonatal haemolysis).

SOURCES CHECKED **[197]** Nitrofurantoin 100 mg Capsules (UK SmPC) **[198]** Nitrofurantoin, oral (US label) **[199]** Nitrofurantoin, oral (US label, Aurobindo Pharma)

Full references p. 432–457